SMARTER BURNS, SAFER FORESTS:

OPTIMIZING PRESCRIBED FIRE ALLOCATION ACROSS THE U.S.

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AGENDA

1 Problem Overview

2 Project Objective

3 Data Sources and Processing

4 Prediction Model: Fire Risk and Associated Costs

5 Optimization Model: Minimize Firefighting and Externality Costs

6 Recommendation and Final Considerations

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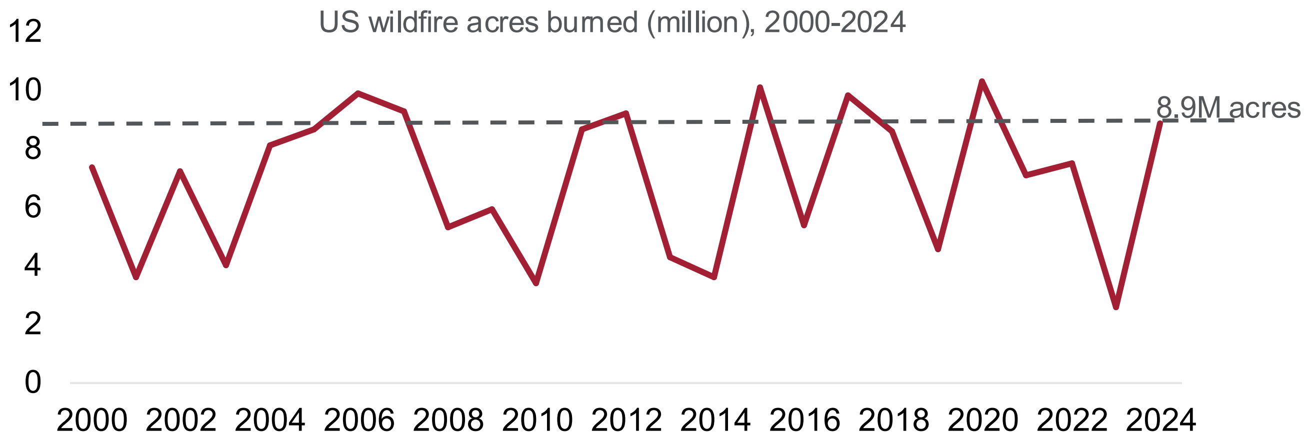
WILDFIRES ARE CATASTROPHIC AND INCREASINGLY COSTLY, BUT PREVENTION EFFORTS FACE CONSTRAINTS



- **Preventative Measure:** Controlled burns of wildfire fuel (e.g. dry shrubs and understory vegetation)
- **Allocation Dilemma:** Smoke externality concerns affect policy and budget
  - West: reactive firefighting vs proactive fuel management
  - Southeast: historical familiarity with timber/land management, controlled burns, smoke externalities

Sources: National Fire Protection Association, National Centers for Environmental Information

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OUR GOAL IS TO MINIMIZE FIREFIGHTING AND SMOKE EXTERNALITY COSTS

Predict

Next-year wildfire likelihood, given 22 years of incident data and 18 features by region

Estimate

Costs of baseline wildfire fighting, smoke externalities, and controlled burns

Optimize

Control burn treatments for minimal costs, given budget and location constraints

Sources: National Fire Protection Association, National Centers for Environmental Information

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HISTORIC WILDFIRE RISK DATA (49 GIGABYTES)

ICS-209-PLUS Hazards	Records of 34,169 US wildfires from 1999-2020
FBFM40 Fuel Models	Vegetation data for entire U.S. (30m resolution)
GHS Urban Centre DB	Urbanized land geometries
WorldPop	Population Density (~1km resolution)
Historic Weather	Monthly high/low temperatures, annual precipitation
US Building Footprints	Locations of 130 million buildings across U.S.

SCOTT & BURGAN FIRE BEHAVIOR FUEL MODEL

45 Primary Fuel Types

- Urban
- Snow/Ice
- Agriculture
- Water
- Barren
- **Short sparse grass** →
- Low load grass
- Moderate load grass
- High load grass
- Dense grass
- Moderate moisture grass
- High load, moist grass
- High load, very coarse
- Dense, tall grass

⋮

10 Fuel Properties

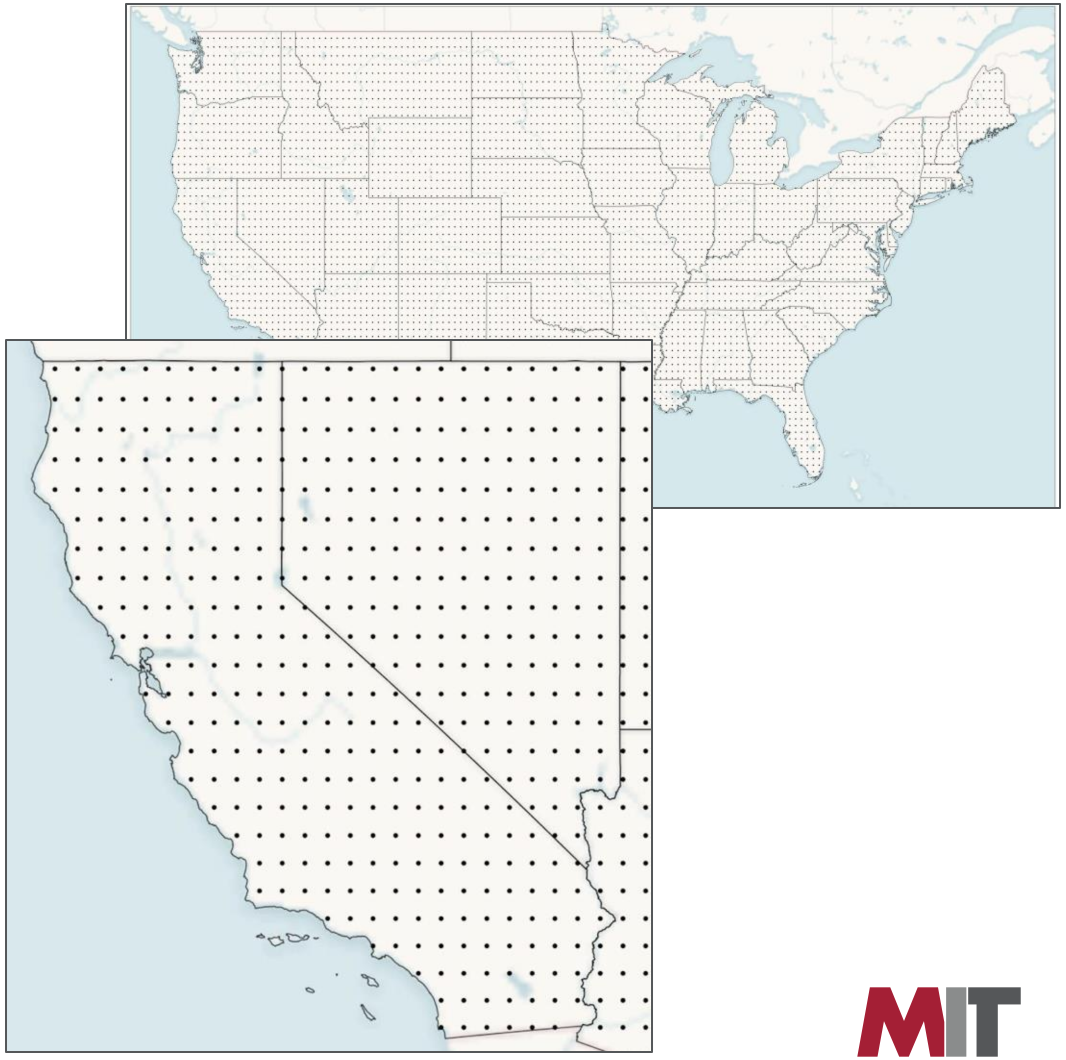
Short sparse grass	Value
1-hr Fuel	0.1
10-hr Fuel	0.0
100-hr Fuel	0.0
Live Herb Fuel	0.3
Live Woody Fuel	0.0
Dead 1-hr Surface Area to Volume Ratio	2200
Live Herb Surface Area to Volume Ratio	2000
Fuel Bed Depth	0.5
Fuel Moisture Content	15
Heat Content	8000

Source: United States Department of Agriculture

FIRE RISK DATA FOR 5,104 CELLS ACROSS US

Processed data for each cell (~ 25 x 25 miles):

- Historic # of Fires
- Average fire size, duration, fighting cost
- 40-Factor Fuel Type Mix
  (e.g. % Dense Grass, % High Load Conifer)
- Fuel Properties
  (e.g. 1/10/100-hr fuel, live herb surface area to volume ratio, fuel bed depth, moisture)
- Monthly High/Low Temperatures
- Annual Precipitation
- Population Density
- Distance to Urban Area
- Structure Density (10/50/100 mi. radius)



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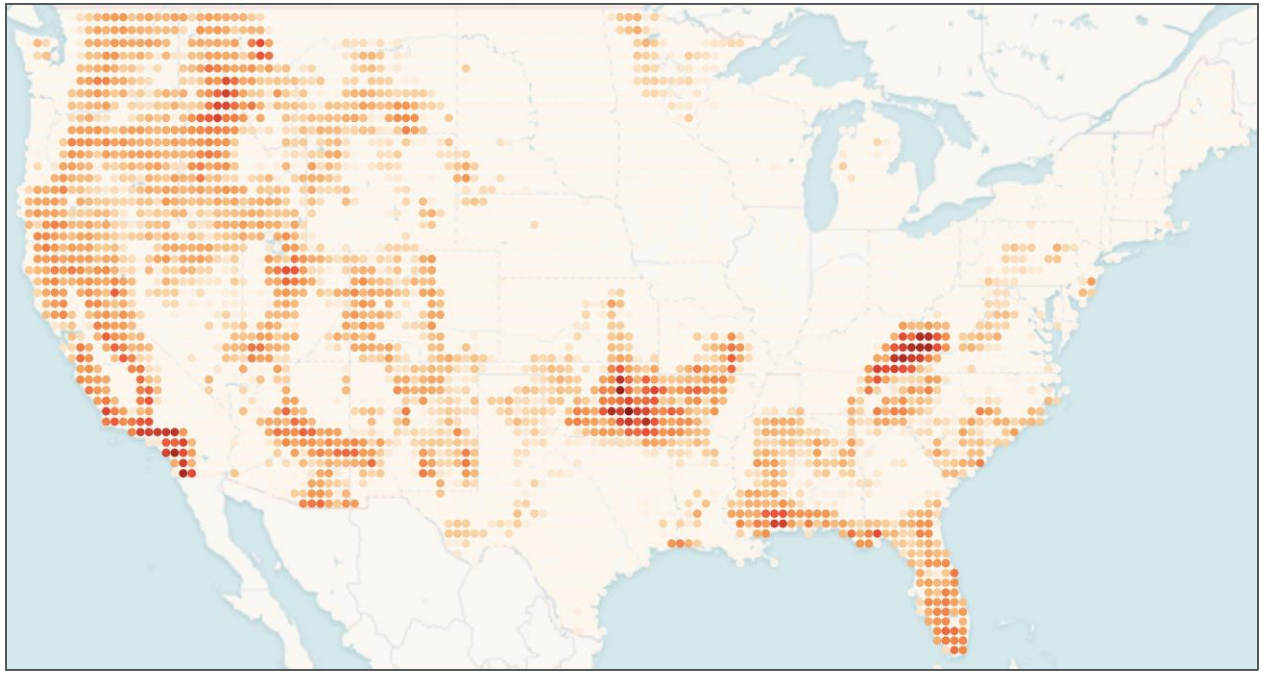
XGBOOST MODEL PREDICTS WILDFIRE RISK

Model: Predict log number of fires, given cell features

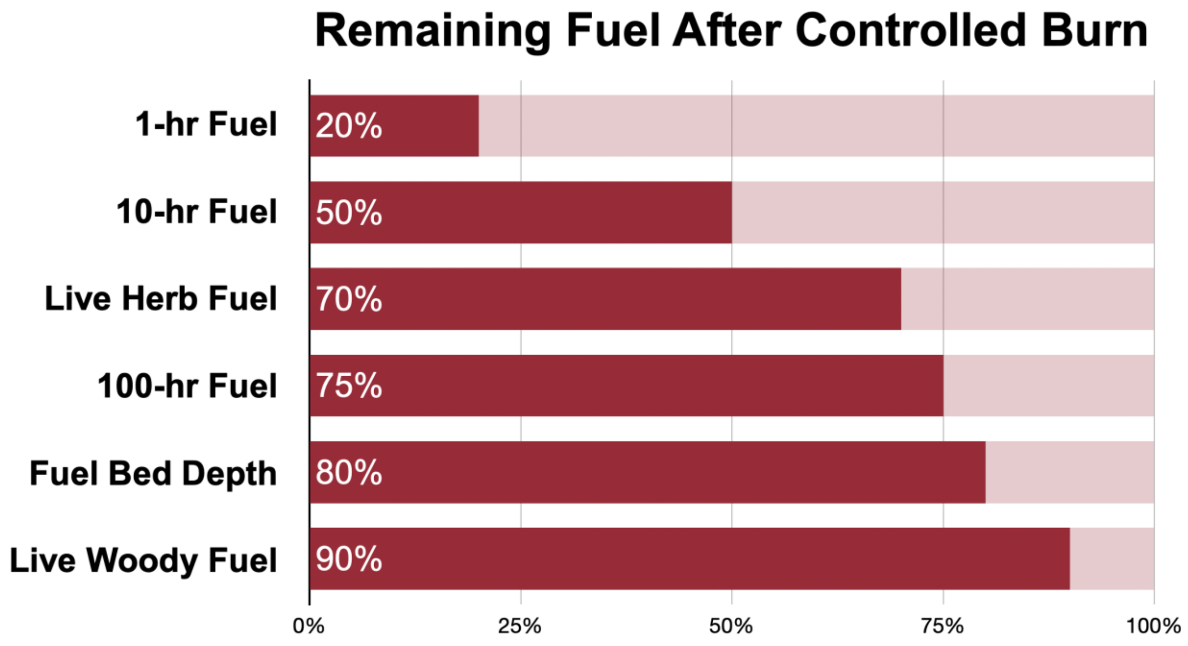
Top Features:

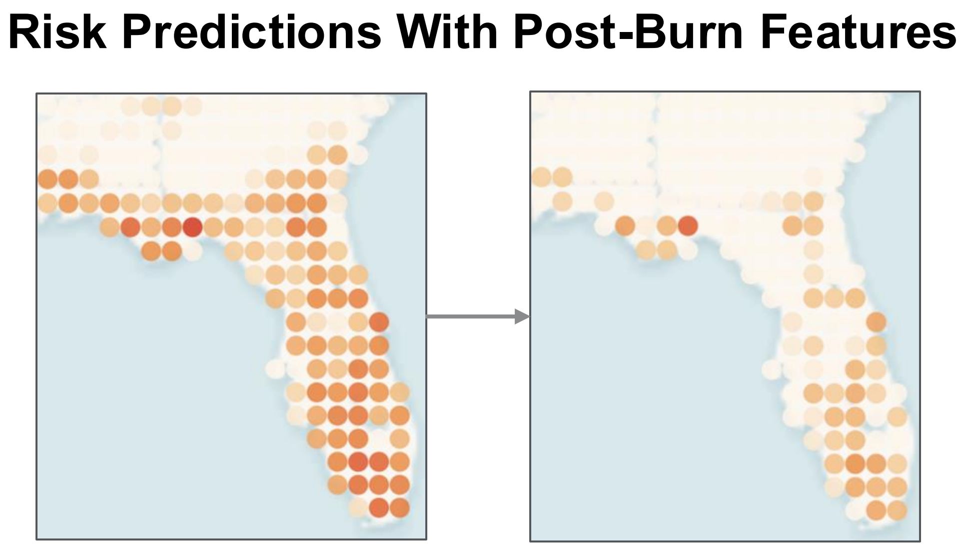
- Fuel Bed Depth
- Live Woody Surface Area/Volume
- 1-hr Fuel Load
- Average Temperature Low
- Annual Precipitation
- Average Temperature High
- Dead 1-hr Surface Area/Volume
- Population

Predicted Wildfire Risk for Each Cell



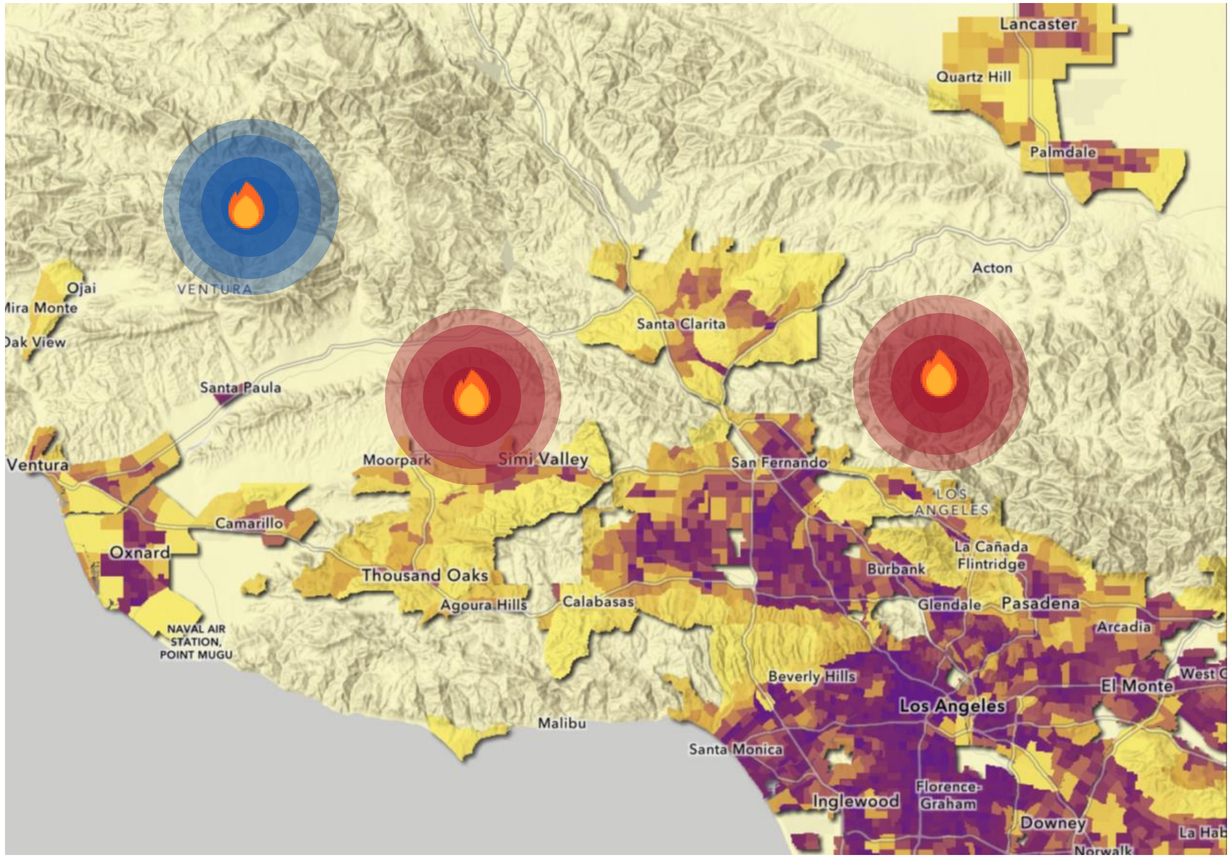
CONTROLLED BURNS REDUCE FIRE RISK





FIRES CLOSER TO DENSE POPULATIONS HAVE HIGHER SMOKE EXTERNALITY HEALTHCARE COSTS

Smoke Externality Model in High-Risk Areas



$$Smoke\ Ext. = \$10 \cdot Days \cdot \sum_{\mathrm{r_{max}}} People \cdot Exposure$$

$\mathrm{r_{max}}$: 1 acre of fire → 700 acres of smoke

$$Exposure(r) = 0.1^{\mathrm{r}/\mathrm{r_{max}}}$$ (10% effect at $\mathrm{r_{max}}$)

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OPTIMIZATION SET-UP: PARAMETERS

Parameters / index		Approach of estimation
$i \in I$	index for grid cells (each cell 0.4 x 0.4 degrees)	N/A
A_{cell}	area of each 0.4 degrees x 0.4 degrees cell (acres)[1]	N/A
$treatable_i$	fraction of cell i that is not urban, water, agriculture, ice/snow or barren	Fuel type mapping
$A_i = treatable_i \cdot A_{cell}$	treatable area in cell i	N/A
F_i	baseline expected 1yr firefighting cost in cell i (before treatment)	Historical average in cell i
F_i'	new expected 1yr year firefighting cost in cell i (after treatment)	Fire likelihood model; post-burn field update model
W_i	baseline expected 1yr fire smoke externality in cell i (before treatment)	Externality model
W_i'	new expected 1yr wildfire smoke externality in cell i (after treatment)	Externality model; post-burn field update model
C_i	operational cost per acre of controlled burn in cell i	Controlled burn cost model
E_i	Smoke externality cost of controlled burn in cell i	Externality model
B^{tot}	Total budget available for operational prescribed-burn cost	Desktop research

1. Calculated precisely based on the latitude/longitude degrees and the bounding box; cell size varies

OPTIMIZATION SET-UP: OBJECTIVE FUNCTION, DECISION VARIABLES AND CONSTRAINTS

Decision Variable	$B_i \geq 0$	acres of controlled burn implemented in cell i
Objective Function	$\min_{B_i} Z = \sum_i [\,(F_i' - F_I) + (W_i{}' - W_i) + E_i]$	Minimize: change in (wildfire firefighting cost + wildfire externality cost) post vs pre-treatment, plus externality costs of controlled burns
Constraints	$\sum_i C_i B_i \leq B^{tot}$	Total operational cost of controlled burn cannot exceed budget
	$0 \leq B_i \leq B_{max,i} \; \forall i$	Per-cell burn limits
	$B_{\max,i} = 10\% \times A_i$ $B_i \equiv 0 \pmod{300}$ $B_i \times \%\,\text{Urban} \leq 10\%$	Burn size needs to be: 1) No larger than 10% of treatable area of cell i 2) In increments 300 acres (typical burn size based on research) 3) 0 if cell has urban density greater than 10%

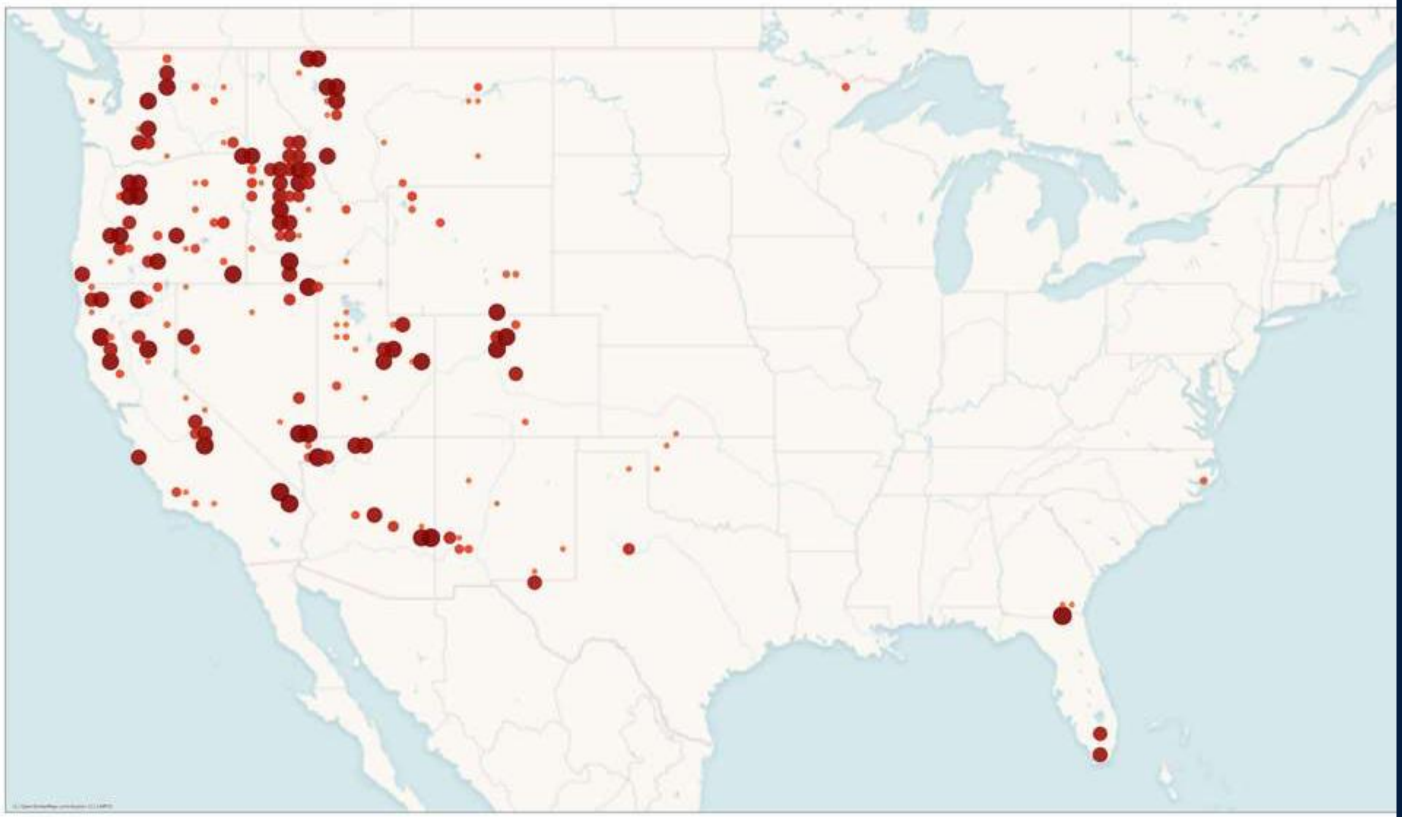
CONTROLLED BURN COST OPTIMIZATION MODEL

Steps	Details
1. Known benchmarks	Average **controlled burn cost / acre of 4 reference landscapes** from literature research**,** indexed by • Great Plains and Grasslands: \$11.37 / acre • Southern States: \$31 / acre • California: \$170 / acre • Idaho: \$194.8 / acre
2. Fuel-mix vectors	Represent each benchmark region and each grid cell as a 40-dimensional fuel vector (we have fuel type mix for each cell; 40 fuel types in total)
3. Similarity weights	For each cell and reference region, define a weight, representing the **cosine similarity** between the two fuel vectors $$w_{i,k} = \frac{T_k \cdot T_{cell,i}}{\parallel T_k \parallel \ \parallel T_{cell,i} \parallel}$$
4. Cost per acre in cell i	**Weighted average** of known regional costs, using the similarities as weights $$C_{burn,i} = \frac{\sum_k w_{i,k} \ K(burn_k)}{\sum_k w_{i,k}}$$

Sources: Great Plains Fire Science Exchange (2024); Forest Service (2016); Resources of the Future (2025); Science Direct (2025)

MODEL'S SUGGESTED CONTROLLED BURNS CREATE AN ESTIMATED $313.6M BENEFIT

Optimization Output – Suggested Controlled Burns Map



Controlled Burn Outputs
- Total operational cost: $299.6M
- Total acres treated: 2,079,000 acres
- Number of burns: 6,930 burns

Externalities & Wildfire Outcomes
- Controlled-burn smoke externality: $110.3M
- Wildfire firefighting cost change: -$22.8M
- Wildfire smoke externality change:-$401.1M

Total Wildfire-related net benefit
- Overall change = -$313.6M

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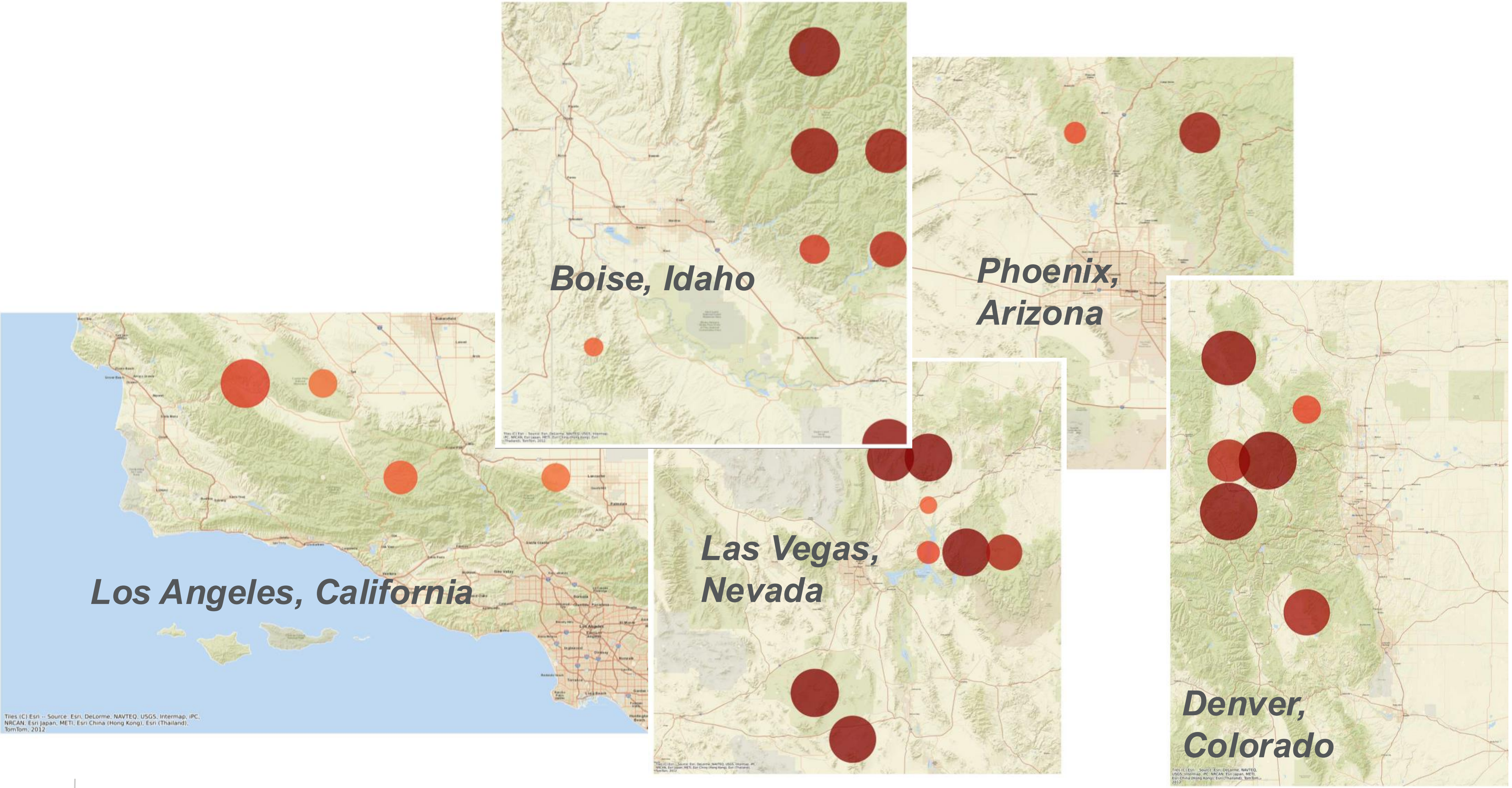
BURNS RECOMMENDED NEAR HIGH-RISK CITIES



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CONCLUSION

Key conclusions

Optimized burns generate strong economic value.
$317.8M in net wildfire-related benefits, including firefighting savings ($23M) and wildfire smoke reduction ($405M), far outweigh controlled-burn operational costs ($300M) and smoke impacts ($110M)

High-impact opportunities are concentrated in the Western US.
Suggested burns cluster around fast-growing WUI[1] regions such as Boise, Los Angeles, Las Vegas, Phoenix, and Denver, revealing clear geographic leverage points for mitigation

Other considerations

Damage cost could not be modeled reliably.
We found no strong predictors of building damage, so we focused on the two components with usable signals: firefighting cost and smoke externalities

Model recommendations differ from real-world burn patterns.
Controlled burns are used far less in the West due to terrain, weather limits, regulations, liability, and lower public tolerance, partially explaining why our optimized locations differ from current practice

1. **WUI (Wildland–Urban Interface):** Areas where homes and infrastructure meet or intermingle with forests, shrublands, or grasslands, regions with the highest wildfire exposure and mitigation priority.

THANK YOU

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